

Screening and Prevention of Diabetes

From guidelines to practice: ADA, EASD, IDF and WHO, with Davidson's and Kumar & Clark's perspectives

Symposium presentation · 25 minutes
[Presenter name, designation, institution]

A growing, largely hidden epidemic

589M

adults (20–79 y) living with diabetes in 2024: 1 in 9

853M

projected by 2050, a 46% increase

43%

of adults with diabetes are undiagnosed (~252 million)

3.4M

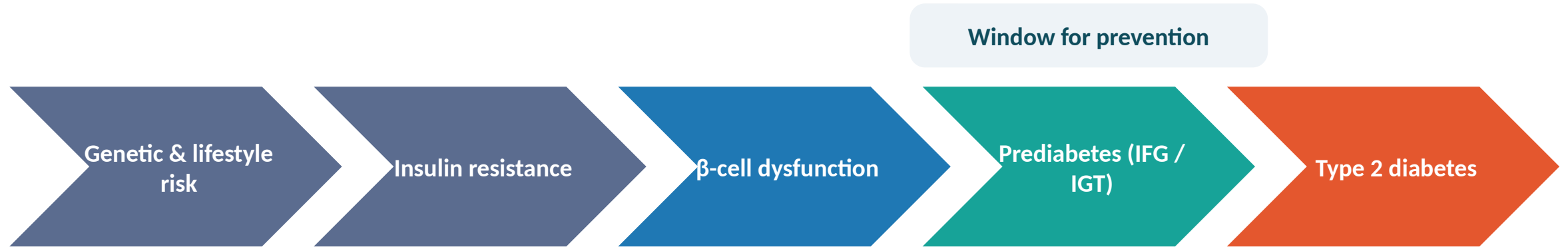
deaths attributed to diabetes in 2024



Why screening matters: type 2 diabetes accounts for >90% of cases and has a long silent phase. Undiagnosed people accumulate micro- and macrovascular damage, and those with prediabetes miss the window in which progression is still reversible.

Source: IDF Diabetes Atlas, 11th edition (2025).

Natural history: the window of opportunity



Hyperglycaemia is typically present for 4–7 years before clinical diagnosis; screening shortens this silent phase.

5–10%

of people with prediabetes progress to diabetes each year

Up to 70%

of people with prediabetes eventually develop diabetes

~50%

already have complications, and β-cell function is ~50% of normal, at diagnosis (UKPDS)

Sources: ADA Standards of Care, Section 3; Tabák et al. Lancet 2012; UKPDS; Davidson's 24e and Kumar & Clark 10e (pathophysiology chapters).

PART 1

Screening

Principles, diagnostic thresholds, who to test, and how

Does type 2 diabetes justify screening?



Important health problem

High prevalence, high morbidity and mortality, rising costs



Long latent phase

Years of asymptomatic hyperglycaemia and prediabetes



Simple, acceptable tests

FPG and HbA1c are cheap, reliable and widely available



Effective early action

Lifestyle and drugs delay diabetes; early control reduces complications



The caveat

ADDITION-Cambridge: population screening of 20,000 adults did not reduce all-cause or CV mortality over ~10 years.

Therefore every guideline recommends targeted (risk-based) screening, not mass population screening, and linking screening to effective follow-up care.

Wilson & Jungner criteria (WHO, 1968); Simmons et al. Lancet 2012 (ADDITION-Cambridge); WHO 2003 consultation on screening for T2D.

Diagnostic thresholds across guidelines

Test	Diabetes (ADA, WHO, IDF, EASD)	Prediabetes: ADA	Intermediate hyperglycaemia: WHO / IDF	High risk: UK / NICE (Davidson, K&C)
Fasting plasma glucose	≥7.0 mmol/L (126 mg/dL)	IFG 5.6–6.9 mmol/L (100–125)	IFG 6.1–6.9 mmol/L (110–125)	5.5–6.9 mmol/L
2-h PG, 75 g OGTT	≥11.1 mmol/L (200 mg/dL)	IGT 7.8–11.0 (140–199)	IGT 7.8–11.0 (140–199)	IGT 7.8–11.0
HbA1c	≥6.5% (48 mmol/mol)	5.7–6.4% (39–47)	Not defined by WHO; ≥6.5% diagnostic (2011)	6.0–6.4% (42–47)
Random PG	≥11.1 mmol/L + classic symptoms	—	—	—
1-h PG, 75 g OGTT (IDF 2024)	≥11.6 mmol/L (209 mg/dL)	—	≥8.6 mmol/L (155 mg/dL)	—

Confirm in asymptomatic people: two abnormal results from the same sample (e.g. FPG + HbA1c) or the same test on two occasions. HbA1c <6.5% does not exclude diabetes diagnosed by glucose (WHO).

ADA Standards of Care, Section 2; WHO/IDF 2006 and WHO 2011 HbA1c report; IDF 1-hour PG position statement 2024; NICE PH38.

ADA: who should be tested?

≥ 35

Test ALL adults from age 35

If results are normal, repeat at least every 3 years; sooner if risk changes.

Any age: BMI ≥ 25 kg/m² (≥ 23 Asian) + ≥ 1 risk factor

- First-degree relative with diabetes
- High-risk race / ethnicity
- History of cardiovascular disease
- Hypertension $\geq 130/80$ mmHg or on therapy
- HDL < 0.9 mmol/L and/or TG > 2.8 mmol/L
- Polycystic ovary syndrome
- Physical inactivity
- Insulin resistance: severe obesity, acanthosis nigricans
- MASLD (fatty liver disease)

Prediabetes

Test every year

Prior GDM

At least every 3 years, lifelong

Other groups

HIV, pancreatitis, steroids, antipsychotics

Children ≥ 10 y / puberty

Overweight + ≥ 1 risk factor

ADA Standards of Care in Diabetes (2025–2026), Section 2, screening criteria for asymptomatic adults. Use the latest annual update.

WHO, IDF, EASD and UK approaches



WHO

- Targeted, opportunistic screening in primary care (PEN / HEARTS-D)
- Prioritise age ≥ 40 , obesity, family history, hypertension, prior GDM
- No mass population screening



IDF

- Three-step model: identify, measure, intervene
- Risk questionnaire first (e.g. FINDRISC), then FPG / OGTT
- 2024: 1-h PG on OGTT for earlier detection of high risk



EASD / ESC

- Screen all patients with CVD: HbA1c + FPG; add OGTT if inconclusive (2019 ESC/EASD)
- General population: risk scores such as FINDRISC (≥ 15 = high risk)



UK: Davidson, Kumar & Clark

- NICE PH38 two-stage approach: validated risk score, then HbA1c or FPG
- Adults ≥ 40 ; from age 25 in South Asian, Chinese and Black groups
- High risk: re-test yearly

WHO HEARTS-D (2020); IDF consensus on prevention of T2D (2007) and 1-h PG statement (2024); ESC/EASD Guidelines 2019; NICE PH38; Davidson's 24e; Kumar & Clark 10e.

Choosing the screening test

Test	Strengths	Limitations
FPG	Cheap, widely available	Needs 8 h fast; misses isolated IGT; day-to-day variability
HbA1c	No fasting; reflects 2–3 months; low variability	Costly; less sensitive; unreliable in some conditions (see right)
75 g OGTT	Most sensitive; detects IGT; standard in pregnancy	Time-consuming; poor reproducibility; patient burden
1-h PG (OGTT)	Detects risk earlier; shorter test (IDF 2024)	Newer; limited adoption and outcome data

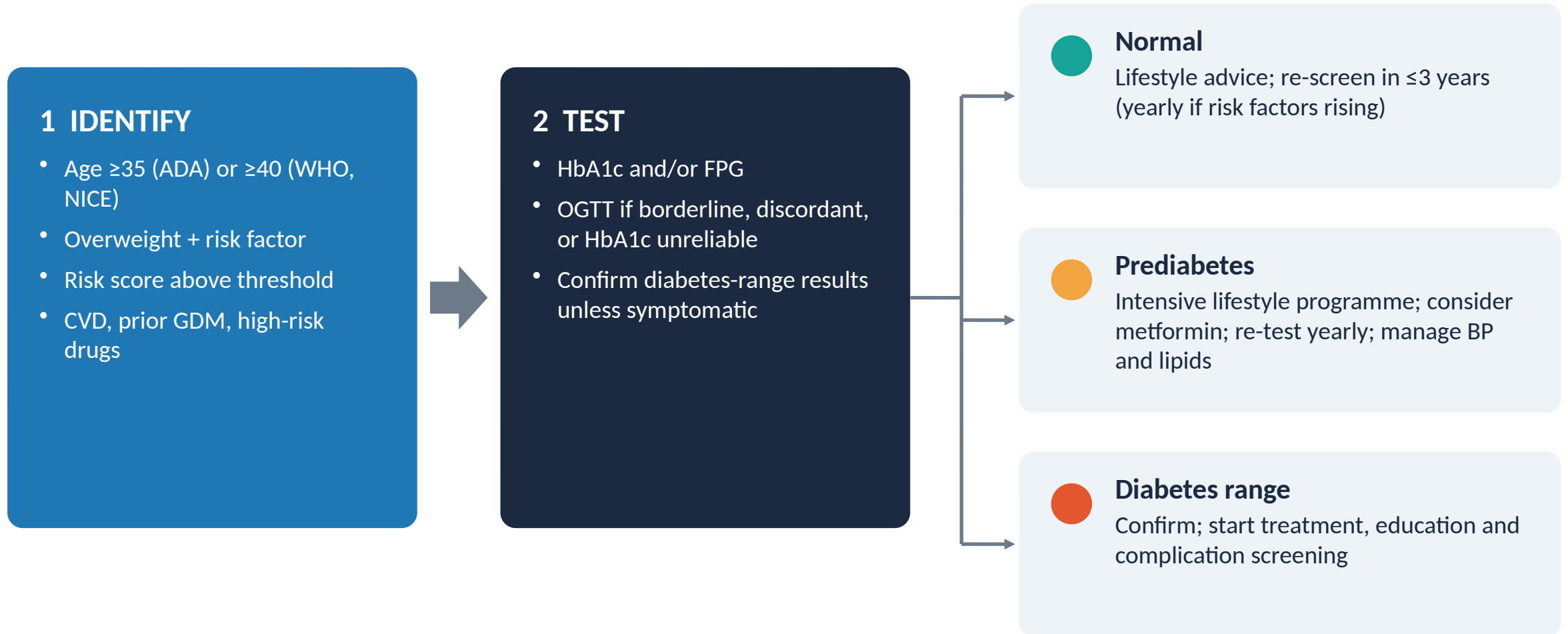
Risk scores before blood tests: ADA Risk Test (≥ 5), FINDRISC (≥ 15), Indian Diabetes Risk Score (≥ 60), Leicester Risk Score (UK).

HbA1c can mislead in...

- Haemoglobinopathies (HbS, HbC, thalassaemia)
- Iron / B12 deficiency, haemolysis, recent bleeding or transfusion
- Pregnancy and the postpartum period
- Advanced CKD, erythropoietin therapy
- HIV on some antiretrovirals
- Rapid-onset (type 1) diabetes, children

ADA Standards, Section 2; IDF 2024; Davidson's 24e (HbA1c limitations); Kumar & Clark 10e.

A practical screening pathway



Synthesised from ADA Standards (Sections 2–3), IDF three-step model, NICE PH38 and WHO HEARTS-D.

Special populations



Pregnancy

Test early in pregnancy if at risk; 75 g OGTT at 24–28 weeks. WHO 2013 / IADPSG: FPG ≥ 5.1 , 1-h ≥ 10.0 , 2-h ≥ 8.5 mmol/L. OGTT 4–12 weeks postpartum, then lifelong screening.



Type 1 diabetes

Islet-autoantibody screening of relatives identifies stage 1 (≥ 2 antibodies, normoglycaemia) and stage 2 (dysglycaemia). Teplizumab delays stage 3 by ~2–3 years (age ≥ 8 , stage 2).



Children & adolescents

ADA: from age 10 or puberty if BMI ≥ 85 th centile plus ≥ 1 risk factor (family history, ethnicity, maternal GDM, insulin resistance); repeat at least every 3 years.



South Asian & other high-risk groups

Diabetes develops ~10 years earlier at lower BMI. Action points: BMI ≥ 23 kg/m²; waist ≥ 90 cm (men) / ≥ 80 cm (women) (IDF). NICE: assess from age 25.

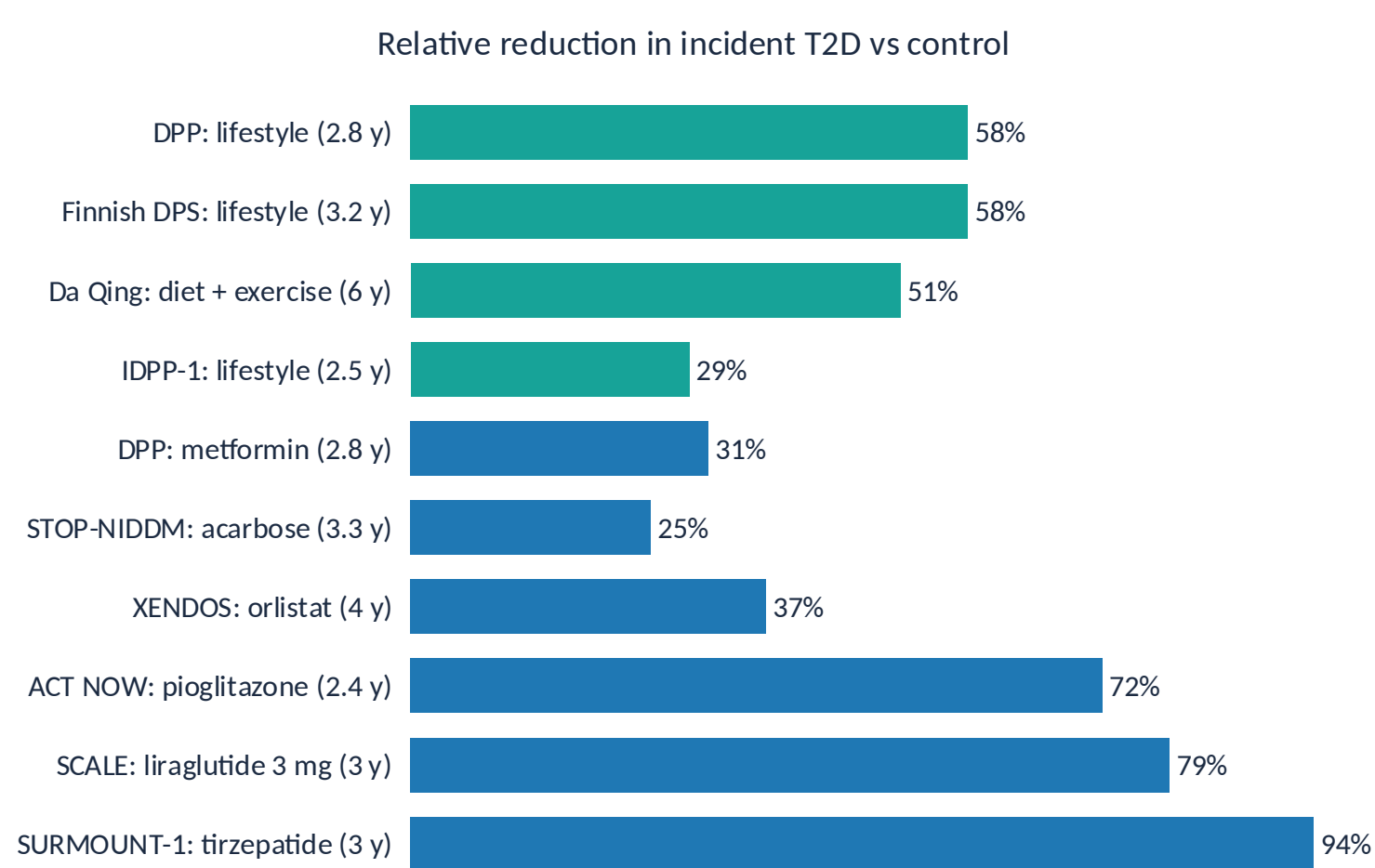
ADA Standards, Sections 2, 14, 15; WHO 2013 hyperglycaemia in pregnancy; 2024 consensus on islet-autoantibody-positive individuals; IDF ethnic waist criteria; NICE PH38.

PART 2

Prevention

Trial evidence, lifestyle, medicines and population policy

Prevention works: landmark trial evidence



Lifestyle

Pharmacological



Lifestyle roughly halves risk

Benefit tracks weight loss: each 1 kg lost $\approx$ 16% lower risk (DPP).



Metformin $\approx$ one-third

Greatest effect with BMI ≥ 35 , age < 60 and prior GDM.



Incretin drugs: largest effect

Driven by weight loss; benefit wanes after stopping.

Trials differ in population, duration and design; not head-to-head. DPP NEJM 2002; DPS NEJM 2001; Da Qing Lancet 2008; IDPP Diabetologia 2006; SURMOUNT-1 NEJM 2025.

Benefits last for decades: the legacy effect

Da Qing, China

30-year follow-up

39%

less diabetes

- CVD events -26%
- All-cause mortality -26%
- Life expectancy +1.4 years
- Diabetes onset delayed ~4 years

DPPOS, USA

15-year follow-up

27%

less diabetes (lifestyle)

- Metformin: 18% reduction
- Benefit persisted after intervention ended
- Metformin cost-saving over time

Finnish DPS

7-year follow-up

43%

less diabetes

- Sustained after counselling stopped
- Benefit proportional to goals achieved
- Basis of the FINDRISC programme

Lifestyle intervention in prediabetes is one of the few interventions shown to reduce mortality (Da Qing).

Gong et al. Lancet Diabetes Endocrinol 2019; DPP Research Group Lancet Diabetes Endocrinol 2015; Lindström et al. Lancet 2006.

The lifestyle prescription



Lose 5–7% body weight

ADA: $\geq 7\%$ initial loss; IDF: 5–10%.
Maintain the loss long term.



Move ≥ 150 min / week

Moderate aerobic activity plus resistance training 2–3×/week; break up sitting every 30 min.



Healthy eating pattern

Mediterranean, DASH or plant-based; whole grains, legumes, nuts and fibre (EASD DNSG 2023, ADA).



Cut sugary drinks

Limit sugar-sweetened beverages, refined carbohydrates and processed meat; moderate alcohol.



Sleep and smoking

Aim for 7–8 h sleep; treat sleep apnoea; stop smoking (raises T2D risk ~ 35 –40%).



Structured programme

DPP model: ≥ 16 sessions in 6 months, then maintenance; group, digital or community delivery.

ADA Standards, Sections 3 & 5; IDF consensus 2007; EASD Diabetes & Nutrition Study Group 2023; Willi et al. JAMA 2007 (smoking).

Pharmacological and surgical options

Option	Key evidence	Guideline position
Metformin	DPP: -31%; IDPP-1: -26%; durable in DPPOS	ADA: consider in prediabetes, esp. age 25–59 with BMI ≥35, FPG ≥6.1, HbA1c ≥6.0%, or prior GDM. NICE: if glucose rises despite lifestyle. Check B12.
GLP-1 RA / tirzepatide	SCALE: -79%; SURMOUNT-1: -94% (3 y)	ADA: obesity pharmacotherapy may be considered in high-risk people with obesity; indication is weight, not diabetes prevention.
Pioglitazone	ACT NOW: -72%; IRIS (post-stroke)	Not routine: weight gain, oedema, fractures, heart failure risk.
Acarbose, orlistat	STOP-NIDDM -25%; XENDOS -37%	Rarely used: GI side effects and poor adherence.
Metabolic surgery	SOS: ~80% lower incidence	For severe obesity; strongest effect on prevention and remission.

Note: in most countries no drug is licensed specifically to prevent diabetes; metformin use is off-label. Lifestyle remains first line in every guideline.

ADA Standards, Section 3; NICE PH38; SCALE Lancet 2017; SURMOUNT-1 NEJM 2025; SOS NEJM 2012; Davidson's 24e; Kumar & Clark 10e.

Beyond the clinic: population prevention

WHO Global Diabetes Compact: 2030 targets



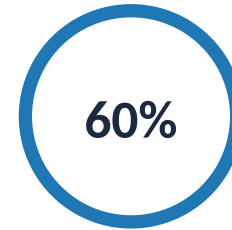
of people with diabetes diagnosed



of those diagnosed with good glycaemic control



with good blood-pressure control



aged ≥ 40 years on statins



with T1D accessing insulin and glucose monitoring



Policy levers (WHO "best buys")

- Taxes on sugar-sweetened beverages
- Front-of-pack labelling; limits on marketing to children
- Healthy school and workplace food; active travel and urban design



Health-system programmes

- NHS Diabetes Prevention Programme (England); CDC National DPP (USA)
- Kerala DPP: peer-led community model in India
- mHealth: SMS coaching cut T2D incidence by 36% in Indian men

WHO Global Diabetes Compact (2021) and WHA75 targets (2022); WHO NCD 'best buys' (2023 update); Ramachandran et al. Lancet Diabetes Endocrinol 2013.

Guidelines at a glance

	ADA	EASD / ESC	IDF	WHO	Davidson • Kumar & Clark (UK)
Who to screen	All ≥35 y; any age if BMI ≥25 (≥23 Asian) + risk factor	All with CVD; risk score (FINDRISC) in general population	Risk questionnaire first, then test high-risk	Targeted, opportunistic in primary care	≥40 y; ≥25 y in high-risk ethnic groups; risk score first
Preferred test	FPG, HbA1c or OGTT	HbA1c + FPG; OGTT if inconclusive	FPG / OGTT; 1-h PG (2024)	FPG / OGTT; HbA1c ≥6.5% acceptable	HbA1c (or FPG)
Intermediate range	FPG 5.6–6.9; HbA1c 5.7–6.4%	IFG / IGT (WHO thresholds)	IFG / IGT; 1-h PG ≥8.6	IFG 6.1–6.9; IGT 7.8–11.0	HbA1c 42–47; FPG 5.5–6.9
Prevention	ILI ≥7% weight loss, 150 min/wk; metformin in selected people	Lifestyle first-line; manage CV risk	Population + high-risk strategy; 5–10% weight loss	Lifestyle plus population policy	Intensive lifestyle; metformin if glucose rises

Glucose values in mmol/L. ILI = intensive lifestyle intervention. Sources as on preceding slides.

Take-home messages

1

Nearly half of people with diabetes are undiagnosed; targeted, risk-based screening is endorsed by all major guidelines.

2

Use a two-step approach: risk score or risk factors, then HbA1c / FPG, with OGTT where HbA1c is unreliable.

3

Know the thresholds: diabetes ≥ 7.0 / ≥ 11.1 mmol/L / ≥ 48 mmol/mol; intermediate ranges differ (ADA vs WHO vs NICE).

4

Intensive lifestyle change with 5–7% weight loss cuts incident T2D by ~50–60%, with a legacy effect on CVD and mortality.

5

Add metformin or obesity pharmacotherapy in selected high-risk people, and pair clinic care with population policy.

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DISCUSSION

Thank you

Questions and discussion